

**'PPServer1' CM Summary****11/23/2022****Revision 0*****ABSTRACT:***

The purpose of this document is to outline the assortment of active CMs in the MUPP C300 controller. As well as listing the CMs themselves, this document also summarises the purpose and some details of the control modules.

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1. Summary of MUPP CMs

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
100_CF_01	Cabinet Cooling Fans	input	Indication state for C300 cabinet cooling fans, output PVFL 100CF01 indicates health HEALTHY/NOT HEALTHY	TRUE	FALSE	FALSE	FALSE	
100_PS_01	C300 Top Power Supply	input	Indication state for C300 Top power supply health, output PVFL 100PS01 indicates health HEALTHY/NOT HEALTHY	TRUE	FALSE	FALSE	FALSE	
100_PS_02	C300 Bottom Power Supply	input	Indication state for C300 bottom power supply health, output PVFL 100PS02 indicates health HEALTHY/NOT HEALTHY	TRUE	FALSE	FALSE	FALSE	
100_TS_01	C300 Cabinet Temp.	input	Indication state of C300 cabinet temperature, output PVFL 100TS01 indicates temp HIGH/NOT HIGH	TRUE	FALSE	FALSE	FALSE	
3WV_143	Three-way valve 143	input/output	Supply Tank 1, Supply tank 2, Feed Pump to ball mill Control of 3WV-143	FALSE	TRUE	FALSE	FALSE	
3WV_368	Three-way valve 368	input/output	CUFT, Supply tank 1, supply tank 2 Control of 3WV-368	FALSE	TRUE	FALSE	FALSE	
3WV_428	Three-way valve 428	input/output	Lamella underflow pump, supply tank 1, supply tank 2 Control of 3WV-428	FALSE	TRUE	FALSE	FALSE	
3WV_572	Three-way valve 572	input/output	Needle tank underflow pump, supply tank 1, supply tank 2 Control of 3WV-572	FALSE	TRUE	FALSE	FALSE	
AG_101	Supply Tank 1 Agitator	output	AG_101 Logic	FALSE	TRUE	FALSE	FALSE	
AG_321	CUFT Agitator	output	AG_321 Logic	FALSE	TRUE	TRUE	FALSE	
AG_621	CSTR1 Agitator	output	AG_621 Logic	FALSE	TRUE	TRUE	FALSE	
AG_641	CSTR2 Agitator	output	AG_641 Logic	FALSE	TRUE	TRUE	FALSE	
AG_661	CSTR3 Agitator	output	AG_661 Logic	FALSE	TRUE	TRUE	FALSE	
AG_FLT_101	Supply Tank 1 Agitator	input	Fault status of AG_101, output PVFL AGFLT101 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
AG_FLT_121	Supply Tank 2 Agitator	input	Fault status of AG_121, output PVFL AGFLT121 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_FLT_221	BMT Agitator	input	Fault status of AG_221, output PVFL AGFLT221 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_FLT_321	CUFT Agitator	input	Fault status of AG_321, output PVFL AGFLT321 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_FLT_621	CSTR1 Agitator	input	Fault status of AG_621, output PVFL AGFLT621 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_FLT_641	CSTR2 Agitator	input	Fault status of AG_641, output PVFL AGFLT641 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_FLT_661	CSTR3 Agitator	input	Fault status of AG_661, output PVFL AGFLT661 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
AG_OFF_121	Supply Tank 2 Agitator	output	AG_121 Logic	FALSE	TRUE	TRUE	FALSE	Inconsistent nomenclature of CMs, should follow AG_101,321,621,641,661
AF_OFF_221	BMT Agitator	output	AG_221 Logic	FALSE	TRUE	TRUE	FALSE	Inconsistent nomenclature of CMs, should follow AG_101,321,621,641,661
AG_RUN_101	Supply Tank 1 Agitator	input	Indication of agitator state output PVFL AGRUN101 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
AG_RUN_121	Supply Tank 2 Agitator	input	Indication of agitator state output PVFL AGRUN121 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
AG_RUN_221	BMT Agitator	input	Indication of agitator state output PVFL AGRUN221 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
AG_RUN_321	CUFT Agitator	input	Indication of agitator state output PVFL AGRUN321 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
AG_RUN_621	CSTR1 Agitator	input	Indication of agitator state output PVFL AGRUN621 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
AG_RUN_641	CSTR2 Agitator	input	Indication of agitator state output PVFL AGRUN641 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
AG_RUN_661	CSTR3 Agitator	input	Indication of agitator state output PVFL AGRUN661 indicates status ACTIVE/INACTIVE	TRUE	FALSE	FALSE	FALSE	
BM_FLT_201	Ball Mill	input	Indication of ball mill fault state output PVFL BMFLT201 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
BM_G1_201	Ball Mill Guard	input	Indication of ball mill guard position (position 1) output PVFL BMG1201 indicates positive position POSITION 1/NO POSITION	TRUE	FALSE	FALSE	FALSE	
BM_G2_201	Ball Mill Guard	input	Indication of ball mill guard position (position 2) output PVFL BMG2201 indicates positive position POSITION 2/NO POSITION	TRUE	FALSE	FALSE	FALSE	Inconsistent nomenclature of PVFLs for two ball mill guard positions
BM_REF_201	Ball Mill	none (logic only)	Ball Mill logic and control	FALSE	TRUE		FALSE	
BM_RUN_201	Ball Mill	input/output	Ball Mill logic and indication of ball mill state output PVFL BMRUN201 indicates status RUNNING/NOT RUNNING	FALSE	TRUE		FALSE	
BM_TANK	Ball Mill Tank	none (logic only)	BMT logic; contains blocks to perform program execution relevant to ball mill page of HMI (304)	FALSE	TRUE		FALSE	part of Joshua Dring's thesis
BM_TANK_OP	Ball Mill Tank	input/output	cycles the valves, motors, and pumps present on pg 303 of the HMI	FALSE	TRUE		TRUE	Automation SCM, part of Joshua Dring's thesis
BMP_FLT_241	Ball Mill Pump	input	indication of fault status of ball mill pump output PVFL BMPFLT241 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
BMP_OFF_241	Ball Mill Pump	input/output	ball mill pump status logic, determines if BMP is on or off	FALSE	TRUE	TRUE	FALSE	Contains configurable interlocks PVs relating to flow out of BMP, BMT level, and CUFT level.
BMP_REF_241-R1A	Ball Mill Pump	none (logic only)	ball mill pump speed reference control. (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
BMP_REF_241_MNTC	Ball Mill Pump	none (logic only)	ball mill pump speed reference control (Alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	**
CA_623	CSTR1?	none (logic only)	unknown (no commenting) conductivity sensors?	FALSE	TRUE	FALSE	FALSE	!!
CA_643	CSTR2?	none (logic only)	unknown (no commenting) conductivity sensors?	FALSE	TRUE	FALSE	FALSE	!!
CA_663	CSTR3?	none (logic only)	unknown (no commenting) conductivity sensors?	FALSE	TRUE	FALSE	FALSE	!!
CM_FF_Test	N/A	N/A	Foundation fieldbus example CM	FALSE	FALSE	FALSE	FALSE	FOUNDATION FIELDBUS EXAMPLE CM

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
CRP_FLT_341	Cyclone Recycle Pump	input	indication of fault status of cyclone recycle pump output PVFL CRPFLT341 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
CRP_OFF_341	Cyclone Recycle Pump	input/output	cyclone recycle pump status logic, determines if CRP is on/off	FALSE	TRUE	TRUE	FALSE	Contains configurable interlocks PVs relating to flow out of CRP, 3WV-368 position, supply tank levels, and CUFT level.
CRP_REF_341-R1A	Cyclone Recycle Pump	none (logic only)	CRP speed reference control. (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
CRP_REF_341_MNTC	Cyclone Recycle Pump	none (logic only)	CRP speed reference control (Alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	**
CSTR_TANKS	CSTR bank	none (logic only)	CSTR bank logic; contains blocks to perform the program execution relevant to CSTR page of HMI (306)	FALSE	TRUE	FALSE	FALSE	part of Joshua Dring's thesis
CSTR_TANKS_OP	CSTR bank	input/output	cycles the valves, motors, and pumps present on pg 306 of the HMI	FALSE	TRUE	FALSE	TRUE	Automation SCM, part of Joshua Dring's thesis
CUF_TANK	Cyclone Underflow Tank	none (logic only)	CUFT logic, contains blocks to perform the program execution relevant to cyclone page of HMI (306)	FALSE	TRUE	FALSE	FALSE	part of Joshua Dring's thesis
CUF_TANK_OP	Cyclone Underflow Tank	input/output	cycles the valves, pumps, and motors present on pg 306 of the HMI	FALSE	TRUE	FALSE	TRUE	Automation SCM, part of Joshua Dring's thesis
CUFP_FLT_361	Cyclone Underflow Pump	input	indication of fault status of CUFP. Output PVFL CUFPLT361 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
CUFP_OFF_361	Cyclone Underflow Pump	input/output	cyclone underflow pump status logic, determines if CUFP is on/off	FALSE	TRUE	TRUE	FALSE	** Contains configurable interlock PVs relating to flow out of CUFP, BMT level, and CUFT level. **possible irrational interlock**
CUFP_REF_361-R1A	Cyclone Underflow Pump	none (logic only)	CUFP speed reference control (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
CUP_REF_361_MNTC	Cyclone Underflow Pump	none (logic only)	CUFP speed reference control (Alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	** inconsistent nomenclature of CUFP vs CUP in other cyclone underflow pump CMs
DP_FLT_611	Dye Pump	input	Indication of fault status of dye pump output PVFL DPFLT611 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
DP_OFF_611	Dye Pump	input/output	dye pump status logic, determines if DP IS ON/OFF	FALSE	TRUE	TRUE	FALSE	Contains configurable interlocks relating to DT>CSTR SVs SV_614,615,616, and flow out of DP.

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
DP_REF_611	Dye Pump	none (logic only)	DP speed reference control	FALSE	TRUE	FALSE	FALSE	Density transmitters
DT_103	Supply Tank 1	input	Data acquisition of density transmitter on supply tank 1	FALSE	TRUE	FALSE	FALSE	Density transmitters
DT_123	Supply Tank 2	input	Data acquisition of density transmitter on supply tank 2	FALSE	TRUE	FALSE	FALSE	Density transmitters
DT_223	Ball Mill Tank	input	Data acquisition of density transmitter on ball mill tank	FALSE	TRUE	FALSE	FALSE	Density transmitters
DT_323	Cyclone Underflow Tank	input	Data acquisition of density transmitter on CUFT	FALSE	TRUE	FALSE	FALSE	Density transmitters
FCV_541-R1A	FCV (NLT input)	output	nonlinear tank input FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_541_MNTC	FCV (NLT input)	output	nonlinear tank input FCV control (alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	** FCVs inconsistent nomenclature with other maintenance mode controls
FCV_570-R1A	FCV (NT to CSTR1)	output	needle tank to cstr1 FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_571-R1A	FCV (NT to STs)	output	Needle tank to supply tanks FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_574-R1A	FCV (NT to BM)	output	needle tank to ball mill FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_622-R1A	FCV (Steam to CSTR1)	output	CSTR1 steam input FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_642-R1A	FCV (Steam to CSTR2)	output	CSTR2 steam input FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_662-R1A	FCV (Steam to CSTR3)	output	CSTR3 steam input FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_688-R1A	FCV (Prod. From PP)	output	product output (from CSTR3 PP) FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FCV_690-R1A	FCV (Reflux from PP)	output	reflux from CSTR3 to CSTR1,2,3 FCV control (Controller MV)	FALSE	TRUE	TRUE	FALSE	Part of 2022 MUPP Controller Revision
FDP_FLT_521	Flow Disturbance Pump	input	indication of fault status of flow disturbance pump output PVFL FDPFLT521 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
FDP_OFF_521	Flow Disturbance Pump	input/output	flow disturbance pump status logic, determines if FDP is on/off	FALSE	TRUE	TRUE	FALSE	contains configurable interlocks relating to flow out of FDP, NT level and NLT level.
FDP_REF_521-R1A	Flow Disturbance Pump	none (logic only)	FDP speed reference control	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
FDP_REF_521_MNTC	Flow Disturbance Pump	none (logic only)	FDP speed reference control (Alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	**

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
FP_FLT_141	Feed Pump	input	indication of fault status of feed pump output PVFL FPFLT141 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
FP_OFF_141	Feed Pump	input/output	feed pump status logic, determines if FP is on/off	FALSE	TRUE	TRUE	FALSE	contains configurable interlocks relating to flow out of FP, supply tank 1 level and supply tank 2 level.
FP_REF_141-R1A	Feed Pump	none (logic only)	FP speed reference control (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
FP_REF_141_MNTC	Feed Pump	none (logic only)	FP speed reference control (Alternative) Maintenance mode	FALSE	TRUE	FALSE	FALSE	**
FT_148	FT (FP to BM)	input	feed pump to ball mill flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_247	FT (BMP to Cyclone)	input	ball mill pump to hydrocyclone flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_347	FT (CRP to BM)	input	cyclone recycle pump to ball mill flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_363	FT (CUFP to STs)	input	cyclone underflow pump to storage tanks flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable !!! Possible mistake on Hopkinson P&ID, unable to determin if FT_363 is CUFP to STs, or CRP to BM
FT_401	FT (Lamella to NT)	input	lamella overflow to needle tank flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_523	FT (FDP to NT)	input	flow disturbance pump to needle tank flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
FT_569	FT (NUFP to CSTR1)	input	needle tank underflow pump to cstr1 flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_573	FT (NUFP to BM)	input	needle tank underflow pump to ball mill flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented
FT_613	FT (DP to CSTR bank)	input	dye pump to cstr bank flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented
FT_687	FT (PP to Output)	input	product pump to output drain flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented, appears to be also doing some kind of conversion from read value to a separate outputted variable
FT_689	FT (PP to CSTR reflux)	input	product pump to cstr reflux piping flow transmitter read	FALSE	TRUE	FALSE	FALSE	uncommented
I/O	N/A	N/A	input/output read/write?	FALSE	TRUE	FALSE	FALSE	
LT_102	LT (Storage Tank 1)	input	storage tank 1 level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_122	LT (Storage Tank 2)	input	storage tank 2 level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_222	LT (BMT)	input	Ball mill tank level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_322	LT (CUFT)	input	cyclone underflow tank level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_422	Bool (LL/HH) Lamella?	input	unknown (no commenting), presumed to be lamella LL and HH sensors. See use in LUP_OFF_421	TRUE	FALSE	FALSE	FALSE	!!
LT_501	LT (NT)	input	needle tank level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_542	LT (NLT)	input	nonlinear tank level transmitter read	FALSE	TRUE	FALSE	FALSE	
LT_667	LT (CSTR3)	input	cstr3 tank level transmitter read	FALSE	TRUE	FALSE	FALSE	
LUP_FLT_421	Lamella underflow pump	input	indication of fault status of lamella underflow pump output PVFL LUPFLT421 indicates FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
LUP_OFF_421	Lamella underflow pump	output	lamella underflow pump status logic, determines if LUP is on/off	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to lamella LL alarm
LUP_REF_421-R1A	Lamella underflow pump	none (logic only)	LUP reference speed control (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
LUP_RUN_421	Lamella underflow pump	input	indication of LUP state output PVFL LUPRUN421 indicates status ON/OFF	TRUE	FALSE	FALSE	FALSE	
MAINTENANCE	N/A	N/A	Part of automated maintenance program, storage of SCM PVs/PVFLs	FALSE	FALSE	FALSE	FALSE	variable storage for SCMs, part of Dring's thesis
MS_100	Maintenance Switch	input	Maintenance switch read, activates maintenance mode output PVFL MS_100 pulses to initiate MM	FALSE	TRUE	FALSE	FALSE	more or less indicate only, activates Dring's maintenance mode
NT_TANKS	NLT/NT	none (logic only)	contains logic to perform execution of program related to NT page	FALSE	TRUE	FALSE	FALSE	part of Joshua Dring's thesis comments say related to cyclone page (305) however logic indicates its related to NT/NLTs
NT_TANKS_OP	NLT/NT	none (logic only)	cycles the valves, pumps, and motors present on og 305 of the HMI	FALSE	TRUE	FALSE	TRUE	Automation SCM, part of Joshua Dring's thesis
NTP_FLT_561	Needle Tank Underflow Pump	input	indication of fault status of needle tank underflow pump output PVFL NTPFLT561 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
NTP_OFF_561	Needle Tank Underflow Pump	input/output	needle tank underflow pump state logic, determines if NTP is on/off	FALSE	TRUE	TRUE	FALSE	**contains configurable interlocks relating to flow from NTP to BMT and CSTR1, NT Level, CSTR3 level **possible irrational interlock**
NTP_REF_561-R1A	Needle Tank Underflow Pump	none (logic only)	needle tank underflow pump reference speed control (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
NTP_REF_561_MNTC	Needle Tank Underflow Pump	none (logic only)	needle tank underflow pump reference speed control (alternative) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	**
OneWireless_CM	N/A	N/A	Handles Wireless comms with Honeywell VIA Modbus TCP/IP	FALSE	TRUE	FALSE	FALSE	wireless comms
OPERATION	N/A	N/A	Uncommented, crucial to plant operation, allows on state of all pumps, sv's, agitators etc.	FALSE	TRUE	FALSE	FALSE	!!
PC3_11_WD1	N/A	N/A	unknown (uncommented)	FALSE	FALSE	FALSE	FALSE	!!! Possibly unused
PID10_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID10_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID11_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID11_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
PID12_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID12_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID13_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID13_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID14_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID14_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID15_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID15_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID16_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID16_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID17_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID17_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID18_PV_SELECT	N/A	N/A	Spare PV routing, not needed for 2022 controller update	FALSE	TRUE	FALSE	FALSE	!!
PID19_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID19_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID1_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID1_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID20_PV_SELECT	N/A	N/A	Spare PV routing, not needed for 2022 controller update	FALSE	TRUE	FALSE	FALSE	!!
PID2_PV_SELECT	N/A	N/A	Spare PV routing, not needed for 2022 controller update	FALSE	TRUE	FALSE	FALSE	!!
PID3_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID3_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID4_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID4_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID5_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
PID5_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID6_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID6_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID7_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID7_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID8_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID8_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID9_PV_SELECT-R1A	N/A	N/A	PV1, PV2, DV routing for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID9_AutVar_SELECT-R0A	N/A	N/A	Checking for if PV1, PV2 selected on HMI is in use by another controller	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PID_PvSp_ARRAY-R1A	N/A	N/A	Stores variable arrays for control system	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PP_FLT_681	Product Pump	input	indication of fault status of product pump output PVFL PPFLT681 indicates fault status FAULT/NO FAULT	TRUE	FALSE	FALSE	FALSE	
PP_OFF_681	Product Pump	input/output	product pump state logic, determines if PP is on/off	FALSE	TRUE	TRUE	FALSE	** contains configurable interlocks relating to flow out of PP via FT_687,689, CSTR3 level, and a combined valve limit imposed on FCV_688,690. **possible irrational interlock**
PP_REF_681-R1A	Product Pump	none (logic only)	Product pump reference speed control (Controller MV)	FALSE	TRUE	FALSE	FALSE	Part of 2022 MUPP Controller Revision
PP_REF_681_MNTC	Product Pump	none (logic only)	Product pump reference speed control (Alternatice) Maintenance Mode	FALSE	TRUE	FALSE	FALSE	**
PPSTLevels1__WD1	N/A	N/A	unknown (uncommented)					!!! Possibly unused
PT_669	Steam Pressure Trans.	input	Data Acquisition for pressure transmitter on steam line	FALSE	TRUE	FALSE	FALSE	
SPARE	N/A	none (logic only)	Labelled as spare PID action control flow	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
ST_TANKS	Storage Tanks	none (logic only)	contains logic to perform execution of program related to Storage tank page (302)	FALSE	TRUE	FALSE	FALSE	part of Joshua Dring's thesis

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
ST_TANKS_OP	Storage Tanks	none (logic only)	cycles the valves, motors and pumps associated with first section of the plant (HMI pp 302)	FALSE	TRUE	FALSE	TRUE	Automation SCM, part of Joshua Dring's thesis
SV_104	Storage Tank 1	output	SV from water addition outlet to storage tank 1	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to storage tank 1 level
SV_105	Storage Tank 1	output	SV from Air addition outlet to storage tank 1	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to storage tank 1 level
SV_124	Storage Tank 2	output	SV from water addition outlet to storage tank 2	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to storage tank 2 level
SV_125	Storage Tank2	output	SV from Air addition outlet to storage tank 2	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to storage tank 2 level
SV_614	CSTR3	output	SV from dye pump to CSTR3	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to CSTR3 level
SV_615	CSTR2	output	SV from dye pump to CSTR2	FALSE	TRUE	TRUE	FALSE	**contains configurable interlock relating to CSTR3 level
SV_616	CSTR1	output	SV from dye pump to CSTR1	FALSE	TRUE	TRUE	FALSE	**contains configurable interlock relating to CSTR3 level
SV_691	CSTR3	output	SV from product pump reflux to CSTR3	FALSE	TRUE	TRUE	FALSE	contains configurable interlock relating to CSTR3 level
SV_692	CSTR2	output	SV from product pump reflux to CSTR2	FALSE	TRUE	TRUE	FALSE	**contains configurable interlock relating to CSTR3 level
SV_693	CSTR1	output	SV from product pump reflux to CSTR1	FALSE	TRUE	TRUE	FALSE	**contains configurable interlock relating to CSTR3 level
SV_AIR	AIR System	output	Determines if AIR supply solenoid is open/closed	FALSE	TRUE	FALSE	FALSE	** Not sure if this SV exists, need to find out
SV_STEAM	STEAM System	output	Determines if STEAM supply solenoid is open/closed	FALSE	TRUE	FALSE	FALSE	** Not sure if this SV exists, need to find out
SV_WATER	WATER System	output	Determines if WATER supply solenoid is open/closed	FALSE	TRUE	FALSE	FALSE	** Not sure if this SV exists, need to find out
TempPC1_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TempPC2_2_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused

General Info				CM information				Notes
CM	Asset Name	channel input/output	CM description	Indicate Only?	Internal Logic?	Interlocks?	SCM?	
TempPC2_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TempPC2OP3_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TempPC2OP_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TmpLevPC4_3_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TmpLevPC5_0_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TmpLevPC6_0_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TmpLvIPC3_WD1	N/A	N/A	unknown (uncommented)	FALSE	TRUE	FALSE	FALSE	!!! Possibly unused
TRBSHT_OP	System Control	none (logic only)	Changes all mode attributes of each device control to program	FALSE	TRUE	FALSE	TRUE	Automation SCM, part of Joshua Dring's thesis
TT_568	CSTR1	input	Data Acquisition from temperature transmitter on CSTR1 feed inlet	FALSE	TRUE	FALSE	FALSE	
TT_623	CSTR1	input	Data Acquisition from temp trans. Inside CSTR1	FALSE	TRUE	FALSE	FALSE	
TT_643	CSTR2	input	Data Acquisition from temp trans. Inside CSTR2	FALSE	TRUE	FALSE	FALSE	
TT_663	CSTR3	input	Data Acquisition from temp trans. Inside CSTR3	FALSE	TRUE	FALSE	FALSE	
WARN_BUZZ	Warning Buzzer	output	Control of warning buzzer	FALSE	TRUE	FALSE	FALSE	** investigate further, reference in Pol's thesis of these being redundant
WARN_LGHT	Warning Light	input	control of warning light allegedly	FALSE	TRUE	FALSE	FALSE	** investigate further, reference in Pol's thesis of these being redundant